



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

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CSEA2

Subject: Programming with Java

ASSIGNMENT No : 7

TITLE : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Outcome

Exercise

1. Create a **Shape application** where Circle and Rectangle inherit from Shape and calculate area.

```
EX1.java    J AS3EX2.java 1    J AS4EX1.java 2    J AS5EX1.java 3    J AS6EX1.java 2    J ShapeDemo.java 3
J ShapeDemo.java > ShapeDemo
1  class Shape {
2  |  void area() {
3  |  |  System.out.println("Area Calculation");
4  |  |  }
5  |  }
6
7  class Circle extends Shape {
8  |  double radius = 5;
9
10 |  void area() {
11 |  |  double a = 3.14 * radius * radius;
12 |  |  System.out.println("Area of Circle = " + a);
13 |  |  }
14 |  }
15
16 class Rectangle extends Shape {
17 |  int length = 10, width = 5;
18
19 |  void area() {
20 |  |  int a = length * width;
21 |  |  System.out.println("Area of Rectangle = " + a);
22 |  |  }
23 |  }
24
25 public class ShapeDemo {
26 |  Run main | Debug main
27 |  public static void main(String[] args) {
28 |  |  Circle c = new Circle();
29 |  |  Rectangle r = new Rectangle();
30 |  |
31 |  |  c.area();
32 |  |  r.area();
33 |  |  }
34 |  }
```

OUTCOME-

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Area of Circle = 78.5
Area of Rectangle = 50
```

2. Develop a **E-Commerce Product System** where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
J Ecommerce.java > Ecommerce
1  interface Product {
2      void display();
3  }
4
5  class ProductDetails {
6      String name;
7      double price;
8
9      ProductDetails(String name, double price) {
10         this.name = name;
11         this.price = price;
12     }
13 }
14
15 class Electronic extends ProductDetails implements Product {
16     Electronic(String name, double price) {
17         super(name, price);
18     }
19
20     public void display() {
21         System.out.println("Electronic Product");
22         System.out.println("Name: " + name);
23         System.out.println("Price: " + price);
24     }
25 }
26
27 class Clothing extends ProductDetails implements Product {
28     Clothing(String name, double price) {
29         super(name, price);
30     }
31
32     public void display() {
33         System.out.println("Clothing Product");
34         System.out.println("Name: " + name);
35         System.out.println("Price: " + price);
36     }
37 }
```

```

36     }
37 }
38
39 class Grocery extends ProductDetails implements Product {
40     Grocery(String name, double price) {
41         super(name, price);
42     }
43
44     public void display() {
45         System.out.println("Grocery Product");
46         System.out.println("Name: " + name);
47         System.out.println("Price: " + price);
48     }
49 }
50
51 public class Ecommerce {
52     Run main | Debug main
53     public static void main(String[] args) {
54
55         Electronic e = new Electronic("Laptop", 50000);
56         Clothing c = new Clothing("T-Shirt", 800);
57         Grocery g = new Grocery("Rice", 1200);
58
59         e.display();
60         System.out.println();
61
62         c.display();
63         System.out.println();
64
65         g.display();
66     }
67 }

```

OUTCOME-

```

PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Electronic Product
Name: Laptop
Price: 50000.0

Clothing Product
Name: T-Shirt
Price: 800.0

Grocery Product
Name: Rice
Price: 1200.0

```

Conclusion

The program was successfully implemented to demonstrate the use of **inheritance** and **interfaces** in Java. It showed how inheritance promotes code reusability by allowing one class to inherit the properties and methods of another, while interfaces provide abstraction and define a common set of behaviors for implementing classes. This practical enhanced the understanding of object-

oriented programming concepts and their role in developing modular, reusable, and maintainable applications.